

LINGHUA ZHU

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RESEARCH INTERESTS

Resource-efficient quantum algorithms, quantum-classical hybrid methods, and quantum machine learning. My research develops adaptive variational algorithms that achieve quantum advantages within hardware constraints, establishes quantum noise-induced regularization demonstrating beneficial noise exploitation in quantum machine learning, and charts systematic pathways from near-term NISQ devices through partial Fault-Tolerance to fully error-corrected systems. Research themes span hardware-aware algorithm design, quantum-ML integration, and strategic noise management, with applications in quantum chemistry, combinatorial optimization, and scientific machine learning.

EDUCATION

New Jersey Institute of Technology & Rutgers University, Newark

August 2012 - May 2018

Ph.D in Physics

Thesis Topic: Theoretical studies of photoinduced dynamics and topological states in materials with strong electron-lattice couplings

University of Science and Technology of China (USTC), Hefei

July 2008 - June 2012

B.S in Physics

Thesis Topic: New generation 2D and 3D solar cells with high conversion efficiency

WORK EXPERIENCE

Sep 2023 - Present

RESEARCH SCIENTIST, University of Washington

Collaborator: Dr. Xiaosong Li

- Established quantum noise-induced regularization theory demonstrating beneficial noise exploitation in quantum machine learning, showing quantum regularization operates at exponentially lower rates than classical dropout.
- Developed partial fault-tolerance frameworks for 25-100 logical qubit regime, characterizing crossover points between NISQ, PFT, and fully error-corrected architectures for quantum chemistry applications.
- Led multi-institution collaboration on quantum computing perspective paper (J. Chem. Theory Comput., accepted with supplementary cover image selection); mentored 2 graduate students and 1 postdoctoral researchers.

Sep 2021 - Aug 2023

POSTDOCTORAL RESEARCH ASSOCIATE, University of Washington

Advisor: Dr. Xiaosong Li

- Created AI-driven shot assignment strategies for variational quantum algorithms, achieving 65-67% measurement reduction while maintaining accuracy across molecular systems.
- Implemented and validated quantum algorithms on IBM and IonQ hardware via Azure Quantum, establishing practical resource requirements for near-term quantum chemistry.
- Applied relativistic quantum chemistry methods to quantum computing for heavy-element molecular simulations.

Oct 2018 - Aug 2021

POSTDOCTORAL RESEARCH ASSOCIATE, Virginia Tech

Advisor: Dr. Sophia Economou & Dr. Edwin Barnes

- Developed ADAPT-QAOA algorithm achieving 60% faster convergence with 50% CNOT gate reduction, published in Physical Review Research with subsequent adoption in high-energy physics and power grid optimization.
- Designed symmetry-preserving quantum circuits reducing variational parameters by $> 50\%$ while improving accuracy by four orders of magnitude, (npj Quantum Information, 2020), adoption in trapped-ion implementations.
- Created adaptive Gibbs state preparation algorithms achieving $> 99\%$ fidelity with theoretical foundations in shortcuts to adiabaticity.

PUBLICATIONS (SELECTED)

First-Author Publications

- **L. Zhu**, S. Liang, C. Yang, and X. Li
Optimizing Shot Assignment in Variational Quantum Eigensolver Measurement
Journal of Chemical Theory and Computation 20(6), 2390–2403 (2024)
- **L. Zhu**, S. Liang, X. Liu, C. Yang, and X. Li
Artificial-Intelligence-Driven Shot Reduction in Quantum Measurement
Chemical Physics Reviews 5, 041403 (2024)
- **L. Zhu**, H.L. Tang, G.S. Barron, N.J. Mayhall, E. Barnes, and S.E. Economou
An Adaptive Quantum Approximate Optimization Algorithm for Solving Combinatorial Problems on A Quantum Computer
Physical Review Research 4, 033029 (2022)
- **L. Zhu**, Y. Dong, and X. Li
Noise-Induced Regularization in Quantum Machine Learning
In preparation (2025)

Co-First-Author Publications (*Equal contribution)

- S. Liang*, **L. Zhu***, X. Li, and C. Yang
QuGStep: Refining Step Size Selection for Gradient-Based Quantum Optimization
APL Computational Physics 1, 026110 (2025)
- B.T. Gard*, **L. Zhu***, G.S. Barron, N.J. Mayhall, S.E. Economou, and E. Barnes
Efficient Symmetry-Preserving State Preparation Circuits for the Variational Quantum Eigensolver Algorithm
npj Quantum Information 6, 10 (2020)
- Y. Chen*, **L. Zhu***, N.J. Mayhall, E. Barnes, and S.E. Economou
How Much Entanglement Do Quantum Optimization Algorithms Require?
arXiv:2205.12283 (2022)
- A. Warren*, **L. Zhu***, N.J. Mayhall, E. Barnes, and S.E. Economou
Adaptive Variational Algorithms for Quantum Gibbs State Preparation
arXiv:2203.12757 (2022)

Major Collaborative Publications

- Y. Alexeev, V.S. Batista, N.P. Bauman, et al. (including **L. Zhu**)
A Perspective on Quantum Computing Applications in Quantum Chemistry using 25–100 Logical Qubits
Journal of Chemical Theory and Computation 2025, 21, 22, 11335–11357,
Selected for supplementary cover image

Application-Focused Quantum Computing

- X. Wei, Z. Zhu, **L. Zhu**, Z. Hu, X. Zhang, G. Wang, S. Bu, and K.W. Chan
Foresight-Seeing Quantum Reinforcement Learning for Two-Stage Unit Commitment with Virtual Power Plants and Renewable Power Integration
arXiv:2410.21240 (2024)
- J. Zhou, Z. Zhu, **L. Zhu**, and S. Bu
Problem-Structure-Informed Quantum Approximate Optimization Algorithm for Large-Scale Unit Commitment with Limited Qubits
iEnergy, vol. 4, no. 4, pp. 215-218, December 2025

- Z. Hu, Z. Zhu, **L. Zhu**, X. Wei, S. Bu, and K.W. Chan
Advancing Hybrid Quantum Neural Network for Alternative Current Optimal Power Flow
arXiv:2410.20275 (2024)
- S. Zhu, Z. Zhu, **L. Zhu**, Y. Ye, S. Bu, and S. Djokic
Evaluating Uncertainties in Electricity Markets via Machine Learning and Quantum Computing
arXiv:2407.16404 (2024)

Earlier Work in Condensed Matter Physics

- K. Qian, **L. Zhu**, K.H. Ahn, and C. Morais Smith
Observation of Flat Frequency Bands at Open Edges and Antiphase Boundary Seams in Topological Mechanical Metamaterials
Physical Review Letters 125, 225501 (2020)
- **L. Zhu**, E. Prodan, and K.H. Ahn
Topologically Protected Flat Bands within Antiphase and Twin Boundaries in Systems with Staggered Interactions
Physical Review B 99, 041117(R) (2019)
- **L. Zhu**, T.F. Seman, M. van Veenendaal, and K.H. Ahn
Photoinduced Nonequilibrium Dynamics in Charge Ordered Materials
Journal of Applied Physics 123, 115112 (2018)

Publication Metrics: 760+ citations (Google Scholar); h-index: 10; i10-index: 11

PRESENTATIONS (SELECTED)

Invited Talks

- **Workshop on Quantum Computing Applications in Quantum Chemistry**
Pacific Northwest National Laboratory & Microsoft, Seattle, WA, April 2025
Chemistry-centric quantum machine learning: From molecular properties to strongly correlated systems
- **Oden Institute Computational Science Seminar**
University of Texas at Austin, Austin, TX, September 2024
Advancements in quantum optimization algorithms: From efficient state preparation to improved convergence
- **SIAM Conference on Optimization (OP21)** (Virtual), 2021
Quantum discrete optimization
- **Benchmarks for Quantum Computing Workshop**
Oak Ridge National Laboratory, Oak Ridge, TN, 2020
An adaptive quantum approximate optimization algorithm for solving combinatorial problems
- **Winter School on Quantum Computing** (Invited Lecture)
Emory University, Atlanta, GA, 2020
Variational quantum algorithms for NISQ devices
- **Practical Quantum Computing (Q2B) Conference**
San Jose, CA, 2019
Resource efficient ansatz for quantum variational algorithms
- **Physics Department Seminar**
Dartmouth College, Hanover, NH, 2019
Designing an efficient circuit for variational quantum eigensolvers

Selected Conference Presentations

- American Physical Society (APS) Global Physics Summit, 2017–2025
Regular presenter of research advances in quantum computing and condensed matter physics

TEACHING & MENTORING

Instructor of Record

- Aug 2012 - May 2015 **Physics Laboratory Courses, New Jersey Institute of Technology, Teaching Assistant**
- Guided undergraduate students through experimental physics labs with individualized support
- March 2022 - May 2023 **EE539 & CHEM 560: Quantum Computing Courses, University of Washington, Instructor**
- Taught quantum computing principles and implementation with hands-on experience using IBM and IonQ quantum hardware via Microsoft Azure Quantum
 - Student course project led to Microsoft Quantum research internship
- Aug 2021 **Quantum Information Science and Engineering High-School Summer School Virginia Tech, Instructor**
- Introduced quantum computing fundamentals using IBM Q Experience to high school students

Research Mentoring

- 2019 - 2021 **Graduate Students (Virginia Tech)**
- Co-mentored Ph.D. student on variational quantum algorithms (2 pubs; joined IBM Quantum ML group)
 - Guided master's student on quantum error correction (admitted to UC San Diego Ph.D. program)
- 2022 - Present **Ph.D Students (University of Washington)**
- Mentored on quantum algorithms, measurement optimization, and quantum-enhanced reinforcement learning
- 2024 - Present **Postdoctoral Researchers (University of Washington)**
- Supervised research on quantum noise-induced regularization and partial fault-tolerance frameworks

SOFTWARE CONTRIBUTIONS

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| <p><i>Flat EB@Topo Bdry</i>
Resulted in two scientific publications</p> <p><i>ADAPT-VQA</i>
Resulted in three scientific publications</p> <p><i>RL-QMeas</i>
Resulted in two scientific publications</p> | <ul style="list-style-type: none"> – Developed method to simulate 2D electronic system in the weak AIII/BDI topology class with topology-structure coupling - Demonstrated the presence of flat zero-mode energy bands in the entire 1D Brillouin zone for states localized within boundaries. – Developed an iterative version of QAOA that is problem-tailored, and which can also be adapted to specific hardware constraints - Speeds up convergence using entangling operators while reducing total number of CNOTs. - Explore how much entanglement is required to speed up optimization algorithms. – Contributed methods for Gibbs state preparation using ADAPT-VQA ansatz - Proposed an easily measured objective function and using dynamically generated algorithm. – Contributed methods for reinforcement-enhanced VQAs - Introduced shot assignment strategies based on estimating standard deviation of measurements. - Trained a reinforcement learning agent towards scalable and resource-frugal VQA. |
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TECHNICAL SKILLS & SOFTWARE

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| <p>Quantum Computing</p> <p>Cloud Platforms</p> <p>Programming & ML</p> | <p>Qiskit, Cirq, PennyLane, OpenFermion, PySCF, Stim</p> <p>AWS Braket, Azure Quantum</p> <p>Python, C/C++, Fortran, PyTorch, TensorFlow, Stable-Baselines3</p> |
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SERVICE

- Journal Reviewer** Physical Review Research (2023–Present); Quantum (2022–Present); Nanoscale Advances (2023–Present); Quantum Science and Technology (2020–Present); New Journal of Physics (2020–Present)
- Organization** Co-organized Virginia Tech Quantum Symposium (2020–2021)
- Two-day event with 20+ participants from Physics, Chemistry, and Computer Science